FitZone Online Class Booking System

Business Analysis Project

Tushar Jain
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Portfolio project. FitZone is a made-up gym that I used to practise the full business analysis process end to end.

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1. Understanding the need

FitZone is a gym in Delhi with around 500 members. Group classes like Zumba, Yoga, HIIT and Spinning are free with the membership, but members have to book a slot because every class has limited space. Right now bookings happen over WhatsApp or in a paper register at the front desk.

I had a first chat with the owner, Rajiv Malhotra, to understand what is going wrong before writing anything formal.

Problem statement

Because bookings are done by hand, popular classes get overbooked (sometimes 20 people for 15 spots), people who book and don't come are never recorded, and the front desk spends a big part of the day sorting out bookings. Members are unhappy and renewals are starting to drop.

Problems

ID	Problem	Who it affects	Impact
P-01	Classes get overbooked because bookings are tracked by hand	Members, trainers	Members get turned away and complain, which leads to fewer renewals
P-02	No proper record of who is coming, so it takes ages to sort out	Front desk, members	Staff can't help members on the floor
P-03	Trainers never know how many people will turn up	Trainers	Class is either too crowded or nearly empty
P-04	Members book and don't show up, and nobody records it	Members, owner	Wasted slot that someone else could have used

Stakeholders

Stakeholder	How they're affected	What they need
Rajiv Malhotra (owner)	Unhappy members don't renew	Fewer complaints, classes used properly
Front desk staff	Most of their time goes on bookings	Members to book themselves
Trainers	Overbooked or empty classes	A list of who's coming before class
Members	Turned away even after booking	Easy booking and a guaranteed spot

What success looks like

- Overbooked classes under 2% within 3 months of launch
- No-shows recorded for every class
- Front desk time on bookings cut by half
- At least 80% of bookings made online by members

(The 50% and 80% targets were my suggestion and Rajiv agreed to them.)

2. Business Requirements Document (BRD)

Project	FitZone Online Class Booking System
Version	1.0
Prepared by	Tushar Jain
Sponsor	Rajiv Malhotra (owner)
Date	16/10/2026
Status	Final

Revision history

Version	Date	Author	Changes
1.0	16/10/2026	Tushar Jain	First draft

Executive summary

FitZone takes class bookings over WhatsApp and a paper register, which leads to overbooked classes, untracked no-shows and a lot of work for the front desk. Members are also comparing us with Cult.fit, where booking is just an app. The plan is an online booking system where members book and cancel their own slots, bookings close once a class is full and trainers mark attendance. The goal is to get overbooking under 2% and halve front-desk booking time within three months.

Background

FitZone has around 500 active members. Classes are free with membership but need a booked slot. This is being raised now because Rajiv rates the current setup 3/10 and renewals have started to drop.

Problem statement

ID	Problem	Impact
P-01	Classes get overbooked because bookings are done by hand	Members turned away, complaints, fewer renewals
P-02	Front desk spends 3-4 hours a shift on bookings	No time for members on the floor
P-03	Trainers don't know how many will come	Can't plan the class
P-04	No-shows aren't recorded	Wasted slots

Business objectives

ID	Objective	Measure	When
BO-01	Reduce overbooking	Under 2% of classes overbooked	3 months after go-live
BO-02	Track attendance	No-shows recorded for every class	From go-live
BO-03	Members book online	80% of bookings made online	3 months after go-live
BO-04	Less work for front desk	Booking time cut by half	3 months after go-live

Scope

In scope	Out of scope
Online booking up to 7 days ahead	Membership payments
Members cancel their own bookings	New member sign-up
Show if a class is full	Personal training
Bookings close automatically when full	Diet / workout plans
Email confirmation	Gym floor management
Trainer booked list and attendance	
No-shows recorded	
Front desk can book for walk-ins	

Stakeholders

Stakeholder	Interest	Responsibility
Rajiv Malhotra (owner)	Fewer complaints	Sponsor, signs off
Front desk	Less booking work	Walk-ins and members who can't book online
Trainers	Know who is coming	Check the list, mark attendance
Members	Easy booking	Book and cancel their own slots

As-Is process

1. Member sends a WhatsApp message to the front desk (or asks at the desk) for a slot.
2. Front desk checks the paper register to see if there's space.
3. Staff reply to confirm and write the name in the register by hand.
4. If the member doesn't turn up, nobody records it and the slot is wasted.

Because everything is done by hand, classes end up overbooked or half empty.

To-Be process

1. Member logs in, sees which slots are free and books one.
2. System sends a confirmation email.
3. System closes bookings when the class is full.
4. Member can cancel online if they can't make it, so the slot opens up.
5. Trainer checks the booked list before class.
6. Trainer marks attendance and no-shows get recorded.
7. Front desk only handles walk-ins.

Business requirements

ID	Requirement	Priority	Source
BR-01	Members must be able to book a class slot online	Must	Owner
BR-02	Members must be able to see if a class is full	Must	Members
BR-03	Members must be able to book up to 7 days ahead	Must	Owner
BR-04	Members must be able to cancel their slot	Must	Members
BR-05	Front desk and trainers must be able to see class details and the booked list	Should	Front desk, trainers
BR-06	Bookings must close automatically when a class is full	Must	Trainers
BR-07	Members must get a confirmation after booking	Must	Members
BR-08	Trainers must be able to mark attendance	Must	Trainers
BR-09	No-shows must be recorded	Must	Owner
BR-10	Front desk must be able to book for walk-ins	Should	Front desk

Assumptions, constraints and dependencies

Type	ID	Description
Assumption	A-01	Members have a smartphone and internet
Assumption	A-02	Trainers will mark attendance at every class
Assumption	A-03	Class timings and limits are set in advance
Constraint	C-01	Budget under Rs. 2 lakh
Constraint	C-02	Live within 3 months
Constraint	C-03	Must work on mobile
Dependency	D-01	Email service for confirmations
Dependency	D-02	Up-to-date member list from the gym
Dependency	D-03	Staff trained before launch

Risks

ID	Risk	Impact	Mitigation
R-01	Members keep using WhatsApp	Medium	Stop taking WhatsApp bookings after launch
R-02	Trainers forget to mark attendance	Medium	Make it a required step at the start of class
R-03	Older members find the app hard	Medium	Front desk can book for them
R-04	System goes down at busy times	High	Pick a reliable vendor with support

KPIs

KPI	Now	Target	Objective
Overbooked classes	~20%	Under 2%	BO-01
No-shows recorded	None	Every class	BO-02
Bookings made online	0%	80%	BO-03
Front desk time on bookings	3-4 hrs/shift	Half	BO-04

Glossary

Term	Meaning
Slot	One place in a class
No-show	Someone who books but doesn't come
Cut-off	Last time you can cancel (2 hours before class)
Walk-in	Member who turns up without booking
Active member	Member with a valid membership

3. Functional Requirements Document (FRD)

Project	FitZone Online Class Booking System
Version	1.0
Prepared by	Tushar Jain
Based on	BRD v1.0
Date	18/10/2026

Purpose

This document explains how the FitZone booking system will work. It turns the business requirements in BRD v1.0 into system functions, rules and screens. It is for the developer, the tester and Rajiv, who approves it.

Scope is the same as in the BRD.

User roles

Role	Who	Can do
Member	Active gym member	Book, cancel and view own bookings
Trainer	Gym trainer	View booked list before class, mark attendance
Front desk	Reception staff	Book and cancel on behalf of members
Admin	Owner or someone he picks	Set up classes, timings and slot limits

Functional requirements

ID	The system shall…	BR	Priority
FR-01	allow a member to book a slot up to 7 days in advance	BR-01, BR-03	Must
FR-02	allow a member to cancel their own booking before the cut-off	BR-04	Must
FR-03	let front desk view class details and booked lists	BR-05	Should
FR-04	let front desk book and cancel for members	BR-10	Should
FR-05	show whether a class is full or how many slots are left	BR-02	Must
FR-06	close bookings when a class reaches its limit	BR-06	Must
FR-07	send a confirmation email after booking	BR-07	Must
FR-08	let trainers view the booked list for their classes	BR-05	Must
FR-09	let trainers mark members present or absent	BR-08	Must
FR-10	record a no-show if the member is absent or not marked by the end of class	BR-09	Must
FR-11	let the admin set up classes, timings and slot limits	BR-06	Must

Business rules

RULE-01: Members can cancel up to 2 hours before the class. After that it's blocked.

RULE-02: Members can book up to 7 days in the future.

RULE-03: One booking per member per class.

RULE-04: Maximum 15 slots per class.

RULE-05: If attendance isn't marked by the end of class, the booking becomes a no-show.

RULE-06: A cancelled slot opens up again for others.

RULE-07: Bookings close when the class starts.

Process flow

See section 4 (To-Be swimlane).

Screens

ID	Screen	Linked to
WF-01	Member login	NFR-02
WF-02	Class schedule	FR-01, FR-05, FR-06
WF-03	Book a slot	FR-01, FR-07
WF-04	My bookings	FR-02, FR-10
WF-05	Trainer attendance	FR-08, FR-09

Full wireframes are in section 5. Front desk and admin screens still need to be designed.

Data requirements

Entity	Field	Type	Mandatory	Rule
Member	Member ID	Number	Y	Unique
Member	Name	Text	Y	Max 50 characters
Member	Mobile number	Text	Y	10 digits
Member	Email	Text	Y	Valid email
Class	Class ID	Number	Y	Unique
Class	Class name	Text	Y	Zumba / Yoga / HIIT / Spinning
Class	Date and time	Date-time	Y	Must be in the future
Class	Slot limit	Number	Y	Max 15
Booking	Booking ID	Number	Y	Unique, system generated
Booking	Member ID	Number	Y	Must be an active member
Booking	Class ID	Number	Y	Must be a scheduled class
Booking	Status	Text	Y	Booked / Cancelled / Attended / No-show

Integrations

Email service only, to send booking confirmations.

Non-functional requirements

ID	Type	Requirement
NFR-01	Performance	Works on mobile and pages load within 3 seconds
NFR-02	Security	Only active members and authorised staff can log in
NFR-03	Availability	Available 99% of the time during gym hours

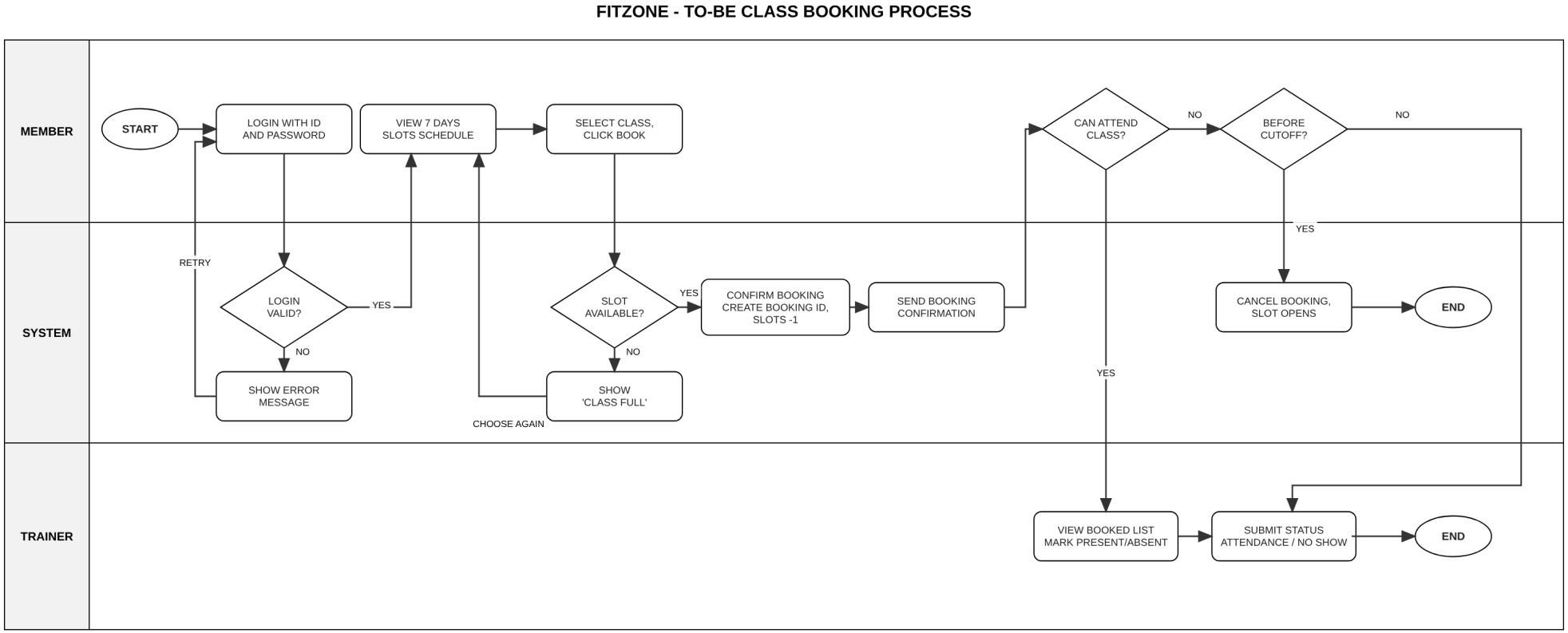
Assumptions and constraints are the same as in the BRD.

Traceability (RTM)

BR	FR	Screen	Test case
BR-01	FR-01	WF-02, WF-03	TC-01
BR-02	FR-05	WF-02	TC-03
BR-03	FR-01	WF-02	TC-02
BR-04	FR-02	WF-04	TC-05, TC-06
BR-05	FR-03, FR-08	WF-05	TC-08
BR-06	FR-06, FR-11	WF-02	TC-03, TC-04
BR-07	FR-07	WF-03	TC-07
BR-08	FR-09	WF-05	TC-09
BR-09	FR-10	WF-04, WF-05	TC-10
BR-10	FR-04	-	-

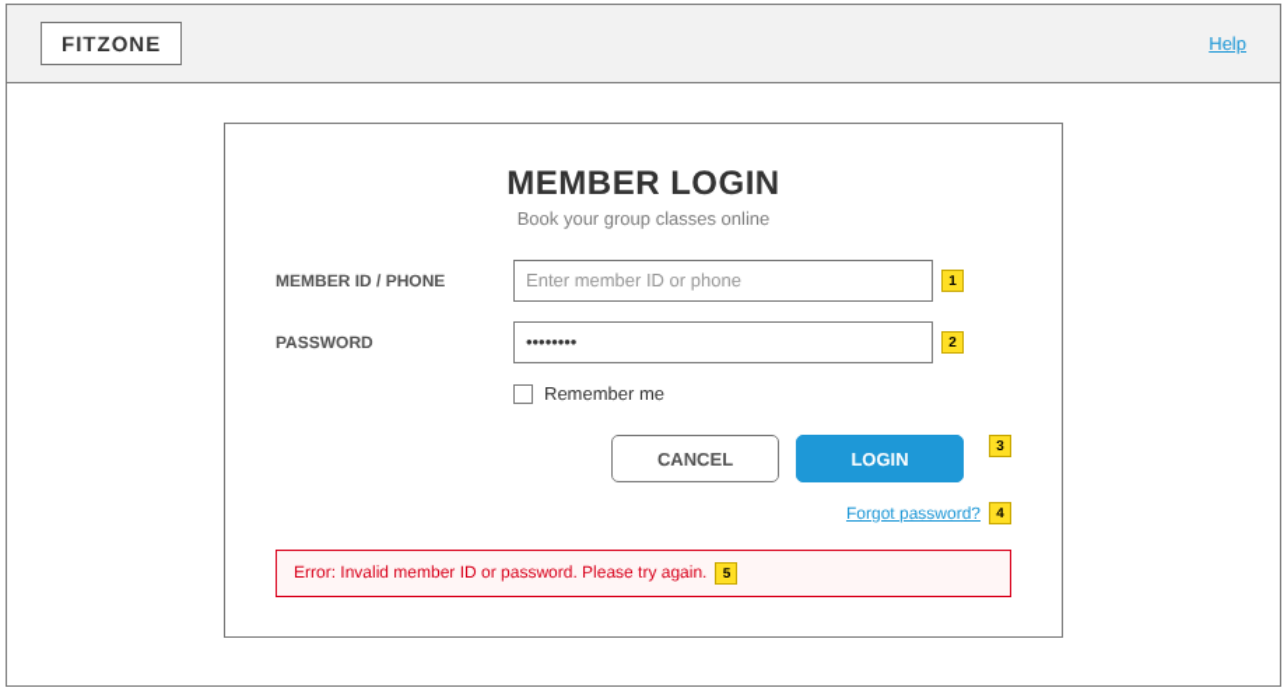
4. Process flow (To-Be)

Made in draw.io. Shows how a booking works between the member, the system and the trainer.



5. Wireframes

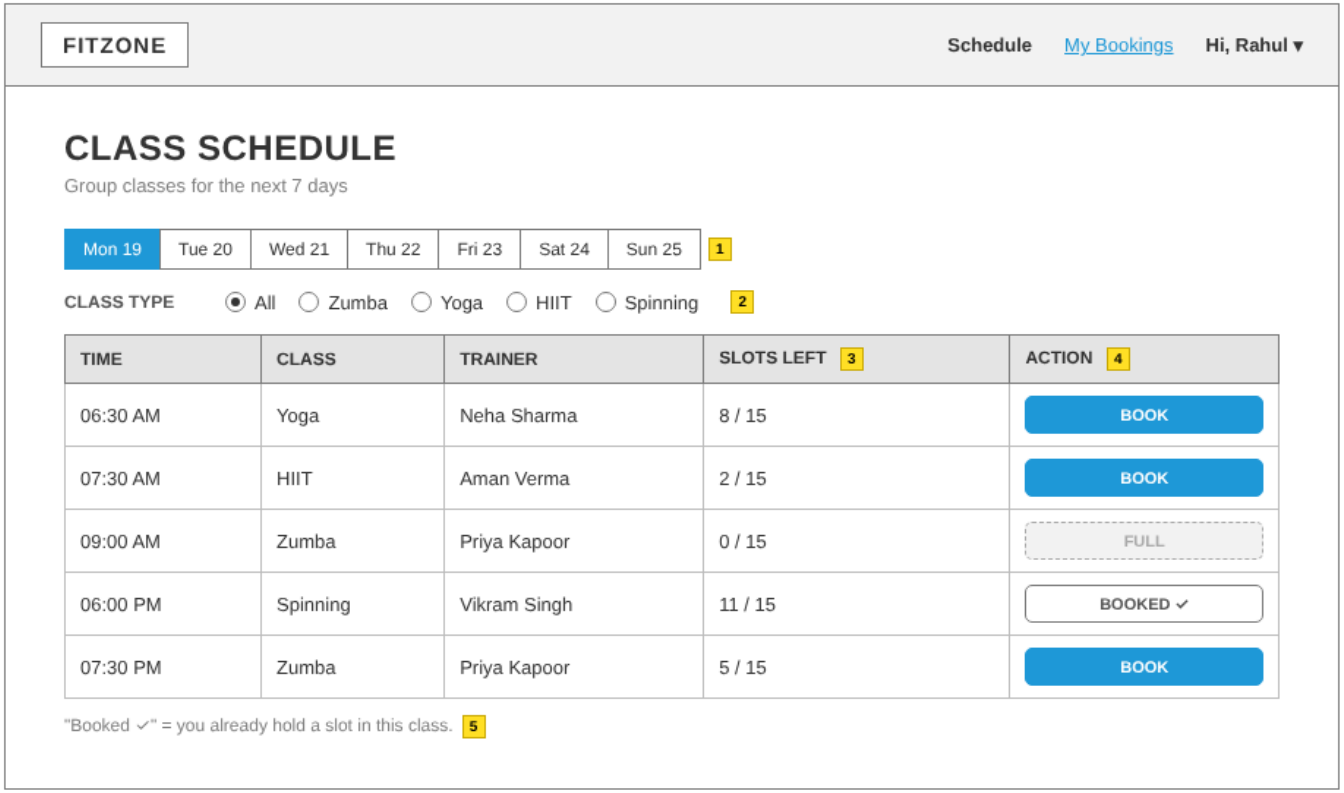
Made in Axure. The yellow numbers match the notes under each screen.



WF-01 · Member Login · FitZone Online Class Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

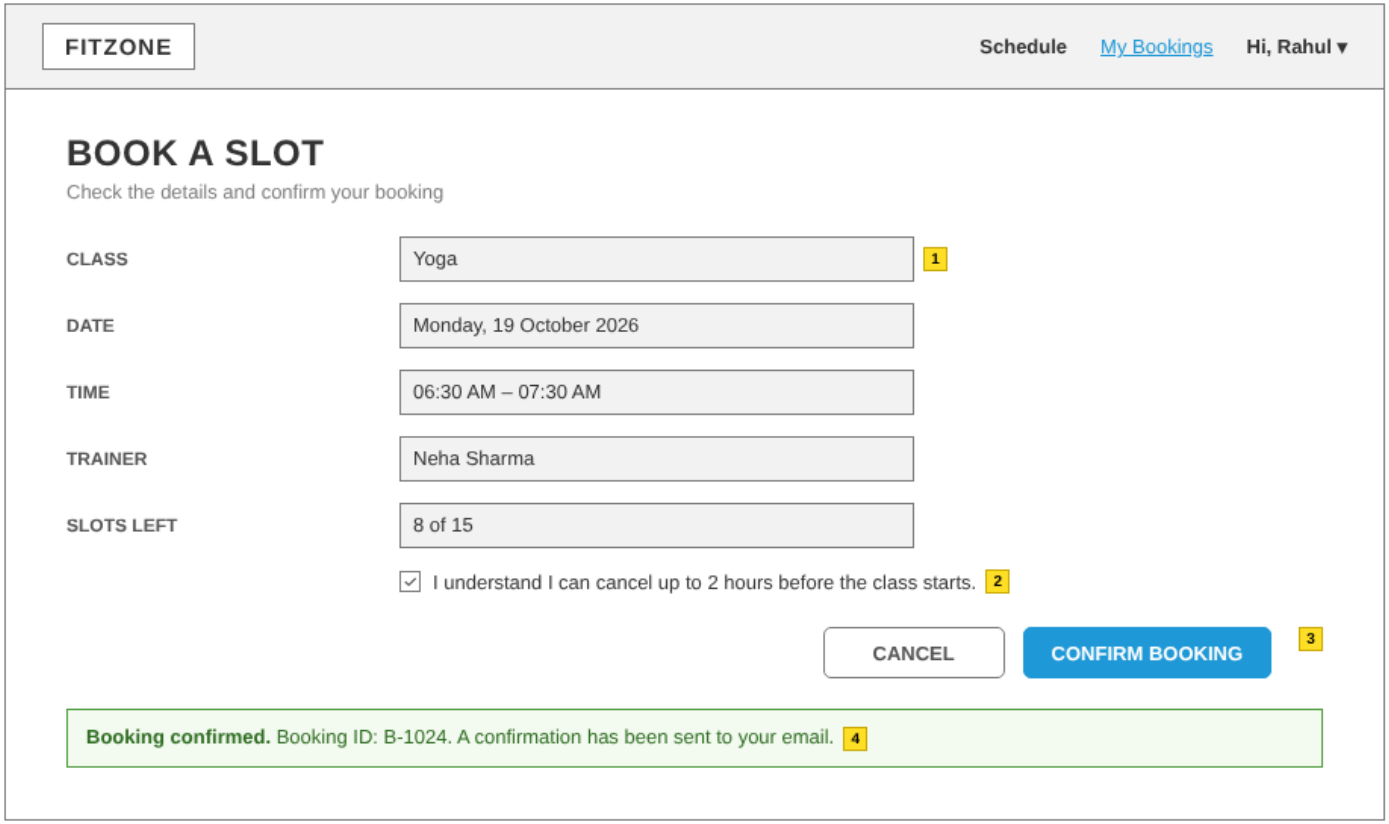
#	WIDGET	NOTE
1	**Member ID / Phone text field**	Member enters their registered member ID or phone number. Mandatory. Only active members can log in.
2	**Password text field**	Member enters their password. Characters are masked. Mandatory.
3	**Login button**	If details are correct, opens Class Schedule (WF-02). If wrong, shows the error message (5).
4	**Forgot password link**	Sends a reset link to the member's registered phone or email. Future scope – not built in this phase.
5	**Error message**	Hidden by default. Shown only when the member ID, phone number or password is wrong.



WF-02 · Class Schedule · FitZone Online Class Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

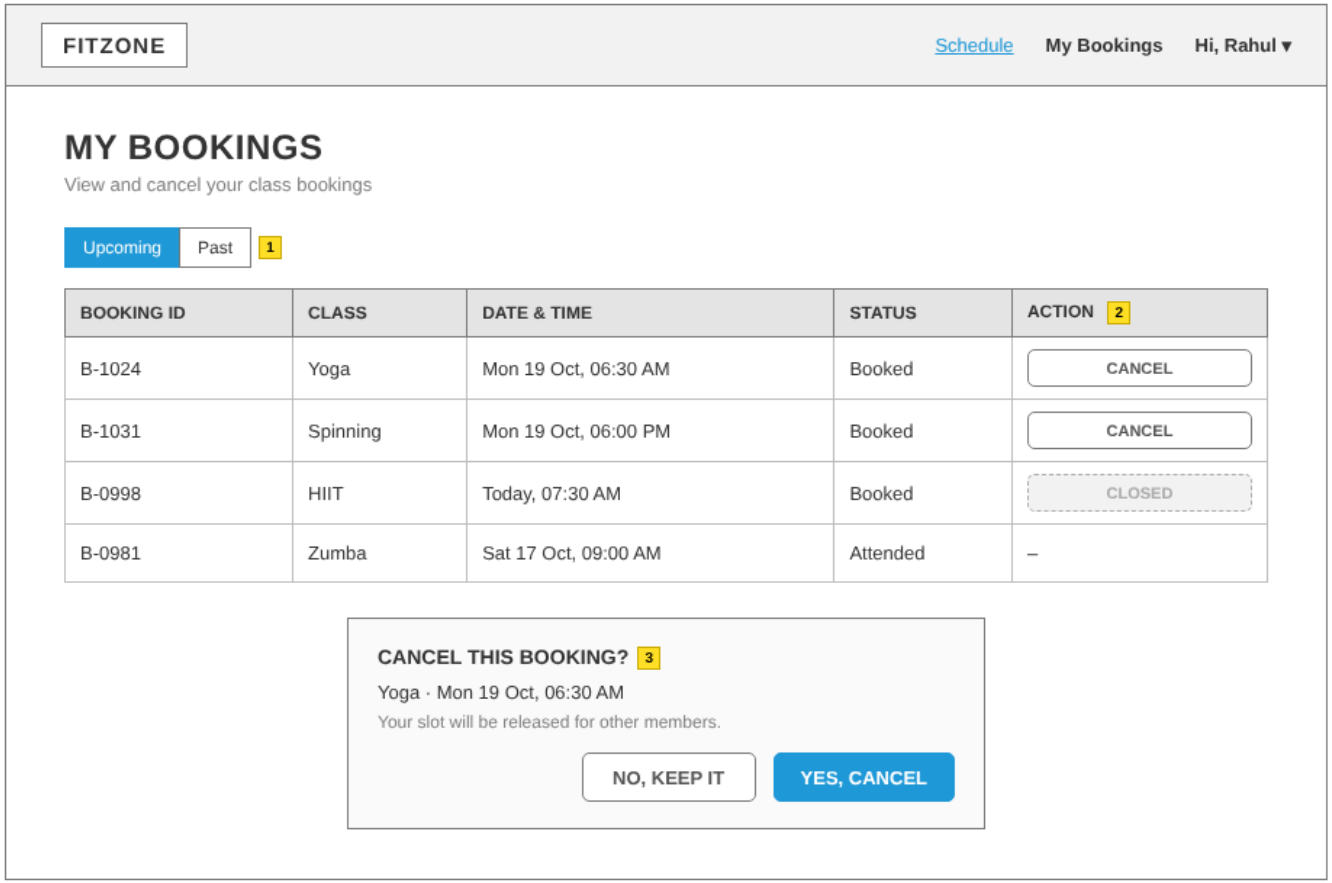
#	WIDGET	NOTE
1	**Date tabs**	Shows the next 7 days only. Members cannot book further ahead (FR-01).
2	**Class type filter**	Filters the list by class. Default is All.
3	**Slots left column**	Live count of free slots out of 15. Updates when someone books or cancels (FR-05).
4	**Book button**	Opens Book a Slot (WF-03). Shows a disabled 'Full' button when no slots are left; bookings close automatically (FR-06).
5	**Booked label**	A member can hold only one booking per class, so Book is replaced by 'Booked ✓'.



WF-03 · Book a Slot · FitZone Online Class Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

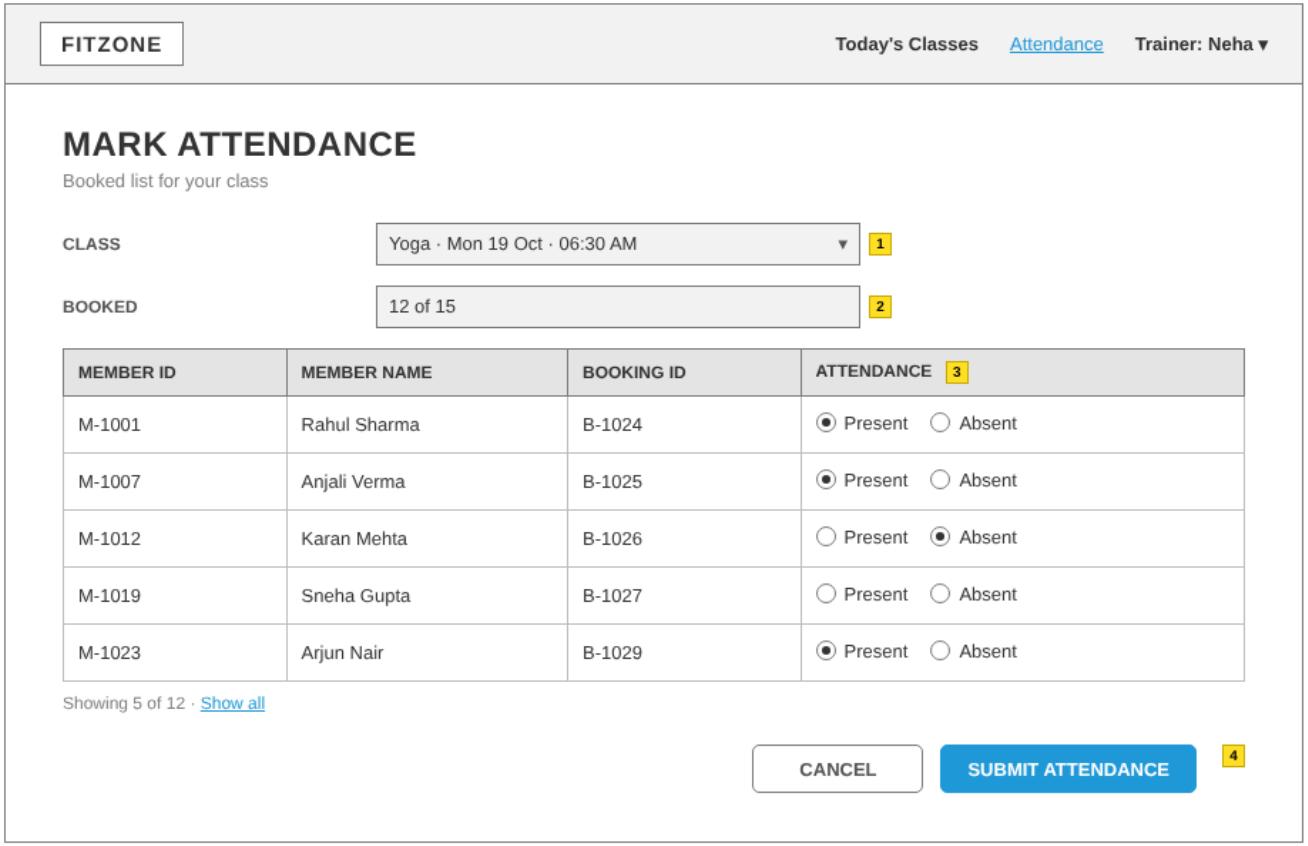
#	WIDGET	NOTE
1	**Class details (read-only)**	Pulled from the selected class on WF-02. Member cannot edit them.
2	**Cancellation checkbox**	Must be ticked before Confirm is enabled. Cut-off of 2 hours to be confirmed with client.
3	**Confirm booking button**	Creates the booking and reduces slots left by 1 (FR-01). If the class filled up in the meantime, shows 'Class is now full'. Cancel returns to WF-02.
4	**Success message**	Hidden by default. Shows the system-generated Booking ID; confirmation email is sent (FR-07).



WF-04 · My Bookings · FitZone Online Class Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

#	WIDGET	NOTE
1	**Upcoming / Past tabs**	Upcoming shows future bookings; Past shows attended and no-show history.
2	**Cancel button**	Member cancels their own booking (FR-02). Shows a disabled 'Closed' button within 2 hours of class start.
3	**Cancel confirmation pop-up**	Hidden by default; opens on Cancel. On 'Yes, cancel' the status becomes Cancelled and the slot reopens for others.



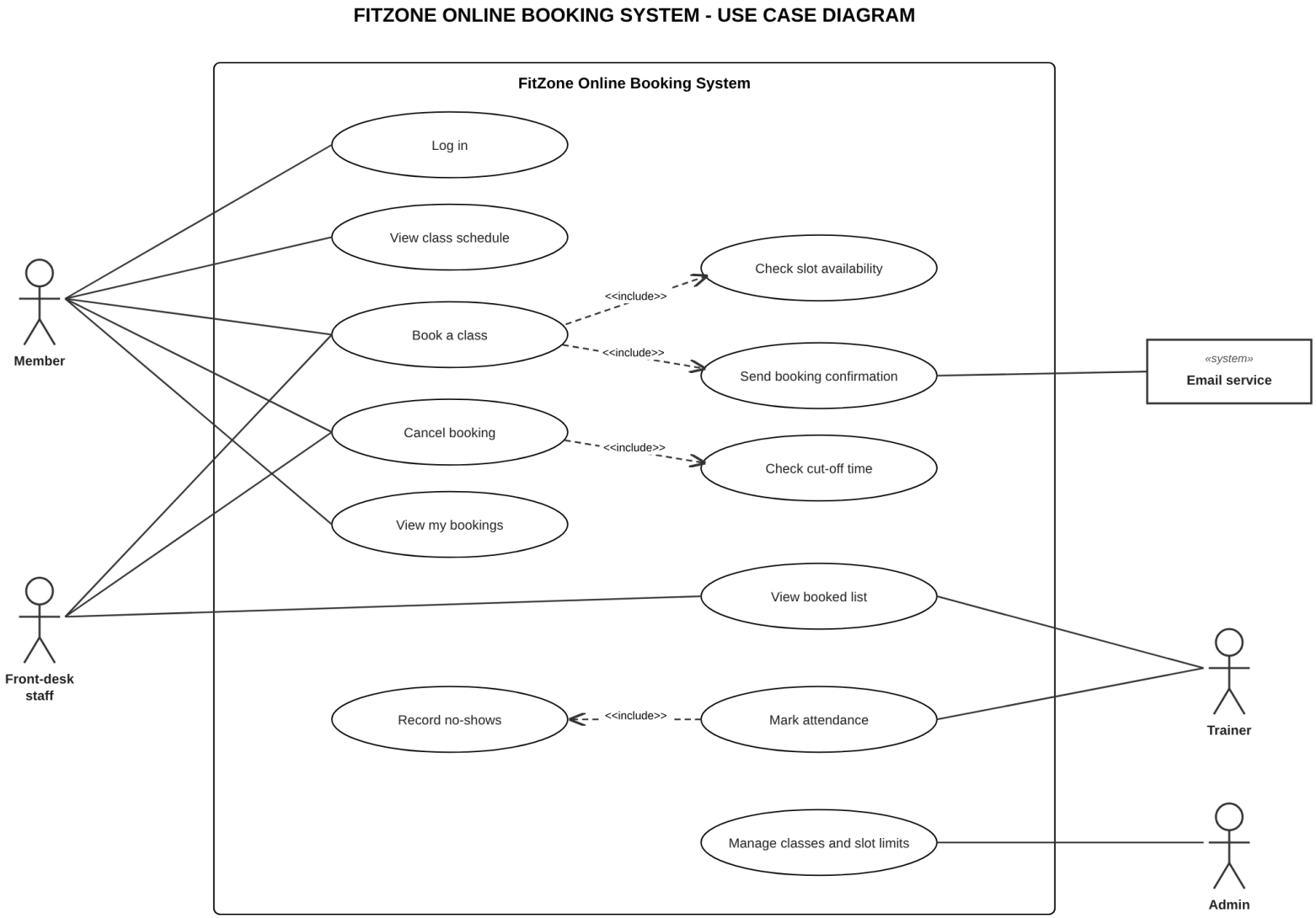
WF-05 · Trainer Attendance · FitZone Online Class Booking · Wireframe v1.0 · Prepared by: Tushar Jain

Widget notes

#	WIDGET	NOTE
1	**Class dropdown**	Shows only the trainer's own classes for today.
2	**Booked count**	Number of members booked against the 15-slot limit, so the trainer knows how many to expect (FR-08).
3	**Present / Absent**	Trainer marks each booked member at the start of class (FR-09). Unmarked rows stay blank until submitted.
4	**Submit attendance button**	Saves attendance. Members marked Absent, or left unmarked when the class ends, are recorded as No-show (FR-10).

6. Use case diagram

Made in draw.io. Shows who uses the system and what they can do.



7. User stories

Written for the backlog in Jira. Story points get estimated by the dev team in sprint planning.

ID	Story	Acceptance criteria	FR
US-01	As a member, I want to book a class online so I don't have to message the front desk	Booking is saved, I get a booking ID and slots go down by 1	FR-01
US-02	As a member, I want to cancel my booking so someone else can have the spot	Works up to 2 hours before class, then the button is disabled	FR-02
US-03	As front desk, I want to see who's booked in a class so I can answer questions	Shows time, trainer, limit and booked names	FR-03
US-04	As front desk, I want to book for walk-ins so they still get a spot	Booking is saved under the member's ID	FR-04
US-05	As a member, I want to see how many spots are left so I don't waste time	Each class shows slots left, full ones say 'Full'	FR-05
US-06	As the owner, I want full classes to close so we never overbook	Can't book when slots left is 0	FR-06
US-07	As a member, I want a confirmation email so I know I'm booked	Email has class, date, time and booking ID	FR-07
US-08	As a trainer, I want to see who's coming so I can plan the class	List and count shown for my own classes	FR-08
US-09	As a trainer, I want to mark attendance so we know who came	Present/absent saved for everyone	FR-09
US-10	As the owner, I want no-shows recorded so I can see who books and doesn't come	Absent or unmarked = no-show in history	FR-10
US-11	As admin, I want to set up classes so the schedule is right	New class shows on the schedule	FR-11

8. UAT test cases

Test cases for user acceptance testing, written from the user stories. These would be run by the owner, front desk and a trainer before go-live.

ID	Test	Steps	Expected result	FR
TC-01	Book a class with free slots	Open schedule, click Book on a class, confirm	Booking confirmed, booking ID shown, slots go down by 1	FR-01
TC-02	Book more than 7 days ahead	Look for dates after the next 7 days	Not shown, can't book	FR-01
TC-03	Book a full class	Open a class with 0 slots left	Shows 'Full', Book button disabled	FR-05, FR-06
TC-04	Two people book the last slot together	Two members confirm at the same time	Only one booking goes through, the other sees 'Class is now full'	FR-06
TC-05	Cancel more than 2 hours before	My bookings, Cancel, Yes	Status Cancelled, slot opens up	FR-02
TC-06	Cancel within 2 hours	Open My bookings	Cancel button disabled	FR-02
TC-07	Confirmation email	Book a class, check inbox	Email with class, date, time and booking ID	FR-07
TC-08	Trainer booked list	Log in as trainer, open today's class	Shows booked count and names, own classes only	FR-08
TC-09	Mark attendance	Mark present/absent, submit	Attendance saved	FR-09
TC-10	No-show	Member marked absent	Shows as no-show in their history	FR-10
TC-11	Expired member login	Log in with expired membership	Access refused	NFR-02